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Vision-Based Closed-Loop Positioning of Motorized Laser Guides for CAD-Referenced Manufacturing

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***Abstract*—Manual measurement and marking remain common in welding, cutting, drilling, and assembly, where task locations defined in computer-aided design (CAD) models must be transferred to the physical workpiece. This paper presents a vision-based closed-loop positioning system for motorized laser guides that projects CAD-defined task geometry onto the workspace. A fixed camera observes a calibration object of known CAD geometry to register image coordinates to the workspace plane, and the task features of a CAD job file in DXF format are mapped into the camera image as targets. A look-and-move visual servoing law then drives stepper-actuated laser projectors until the observed patterns coincide with their targets, so that the controlled variable is the projected pattern itself rather than the motor position. A Raspberry Pi 5 performs image processing, registration, and high-level control, and an STM32F4 microcontroller generates the real-time STEP/DIR signals. An analysis of the quantized control law yields a convergence condition under sensitivity mismatch and a lower bound on the achievable tolerance. In 240 positioning runs on a prototype with two motorized line lasers and tolerances of 4 and 6 mm, every closed-loop run met its tolerance within two moves, and each correction move reduced the error to at most 1.64 mm. Within the closed-loop runs, feedback reduced the root mean square error with respect to the registered targets from 4.53 mm after the first move to 2.22 mm (51%), against 6.57 mm in open loop. Because the calibration plate lies 6 mm in front of the wall, the registration adds a scale error of about 1.2%; accounting for it, the estimated root mean square error on the wall is 3.9 mm in closed loop and 8.6 mm in open loop.**

Index Terms—Visual servoing, laser guidance, CAD-based registration, closed-loop positioning, stepper motor control, embedded control.

I. INTRODUCTION

Many manufacturing operations, including welding, cutting, drilling, and assembly, require locations defined in computer-aided design (CAD) models to be transferred to the workpiece. At manual and small-batch workstations, this is still commonly done by measuring and marking with rulers, templates, and scribes, which is time-consuming and makes the accuracy dependent on operator skill. Projection-based spatial augmented reality displays such information directly on the work surface and is one of the main display modalities for augmented-reality instruction in manual assembly [1]; in industrial site assembly, dynamic in-situ projection improved efficiency, usability, and cognitive workload compared with instructions at a terminal [2]. Laser projectors do the same in production, for example in the layup of large composite parts [3].

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The value of projected guidance depends on how accurately the projection coincides with its intended location. In open-loop projection, errors in the projector model, its mounting, or its registration to the workpiece appear directly in the projected geometry. Recent work reduces them by calibration, fusing binocular vision and laser ranging to relate the camera, galvanometer, and workpiece frames [4]; there, the pattern is measured during calibration and the projection is corrected for each workpiece pose, whereas here the pattern itself is measured and corrected at every positioning iteration. Stepper-actuated laser modules are a low-cost alternative to galvanometer projectors, but their open-loop positioning is further affected by missed steps, the limited accuracy of microstepping under load, transmission backlash, and uncertain mounting geometry; feedback at the motor improves the accuracy of microstepping drives [5].

Visual servoing addresses such modeling errors by closing the loop through camera measurements [6], [7]. When the camera observes both the target and the controlled element, an endpoint closed-loop configuration, the accuracy becomes insensitive to actuator-to-camera calibration errors [6], and convergence requires only an approximate estimate of the sensitivity between actuator motion and image features, as uncalibrated schemes that estimate it online show [8]. Camera feedback has also been applied to laser steering: in vision-guided laser surgery, a spot deflected by an actuated mirror is steered along paths under visual servoing [9], and a laser carried by a flexible tool on a robot arm scans planned trajectories under image-based control [10]. The actuation considered here differs in being quantized and operated in look-and-move fashion, with targets derived from a CAD job file rather than specified in the image. Registering a camera to a planar workspace, which is needed to express CAD-defined locations in the image, is well established through planar homographies and camera calibration [11], [12].

These results suggest treating the projected pattern itself, rather than the motor position, as the controlled variable: the camera then observes the location of the guide on the workpiece, and actuator errors are corrected by feedback instead of calibration. This paper presents such a vision-based closed-loop positioning system for motorized laser guides. Its contributions are: (i) a registration and target-generation pipeline that links CAD job data to image-space targets through a calibration object; (ii) a quantized look-and-move control law, with a convergence condition under sensitivity mismatch and a lower bound on the achievable tolerance; and (iii) an implementation that combines vision processing on a Raspberry

Manuscript 91 submitted to 2026 14th International Conference on Systems and Control (ICSC). Received September 29, 2026.

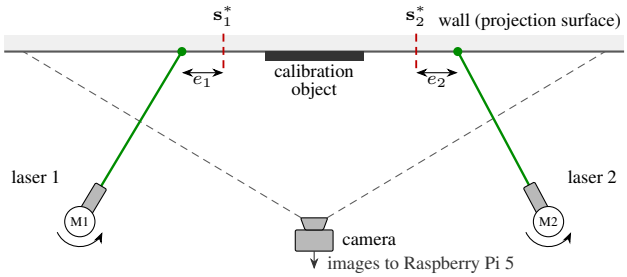


Fig. 1. Top view of the laser guidance system. Each line laser is rotated about the vertical axis of its stepper motor (M1, M2), which moves its projected line (green dot) along the wall; the camera between the lasers observes the lines, their targets $\mathbf{s}_i^*$, and the calibration object.

Pi 5 with deterministic step generation on an STM32F4, evaluated experimentally against open-loop positioning on a prototype with two motorized line lasers.

II. System Description and Data Collection

A. Experimental Laser Guidance System

The experimental platform, a prototype laser guidance system, is shown in Figs. 1 and 2, and its main parameters are listed in Table I. The back panel of an enclosure, the wall, serves as the workspace, onto which two green line lasers project vertical guide lines. Each laser is mounted directly on the shaft of a stepper motor with its own driver, without gearing, so that rotating the laser about the vertical motor axis moves its line along the wall. A fixed camera between the two lasers faces the wall and observes the calibration object, a black rectangular plate, and the projected lines.

Computation is divided between two processors according to their timing requirements. A Raspberry Pi 5 acquires the camera images and performs image processing, geometric registration, CAD interpretation, and high-level control. Pulse generation is delegated to an STM32F4 microcontroller, which receives motion commands from the Raspberry Pi over a UART link and generates the STEP/DIR signals for the stepper drivers.

The second processor provides determinism rather than throughput. A stepper motor follows the commanded motion only while the interval between STEP edges matches the acceleration it can track; a stretched interval acts as an acceleration transient and can cause loss of synchronism. On single-board computers, worst-case latencies of several hundred microseconds have been measured under a standard Linux kernel, reduced but not removed by a real-time preemption patch [13]; at 1000 pulses/s, this is a large fraction of the 1 ms pulse period, and the competing load is the vision pipeline itself. A dedicated microcontroller without a general-purpose operating system or vision workload keeps the actuation timing independent of this latency [14]; the loop would correct a lost step, but only with a further iteration.

B. Experimental Procedure and Data Collection

Each experiment has three phases: calibration, in which the image-to-workspace mapping is estimated from the calibration object (Section III-A); job registration, in which the task features of a CAD job file, such as a welding path, cutting line, circular cutout, drilling position, or assembly boundary, are converted into image targets (Section III-B); and positioning, in which each laser is driven to its target by a single model-based move (open loop) or by the controller of Section III-D (closed loop).

TABLE I
System and Controller Parameters

Parameter	Value
Camera; resolution; frame rate	USB camera; 640×480 px; 30 fps
Camera–wall distance; distortion	$\approx$500 mm; not corrected
Calibration plate	$207 \times 134 \times 6$ mm
Step limit $\Delta q_{\max}$	none (no saturation)
ROI x (laser 1; 2); y	32–307; 332–608; 24–384 px
Laser modules	Two green line lasers
Stepper motors; drivers	NEMA-17; TMC2209, one per motor
Microstepping; transmission	4 per full step; direct drive
STEP frequency	$\approx$1000 pulses/s
Loop gain λ; estimate $\hat{J}$	1; constant (mm to microsteps)
Settling time T_s	1.5 s
Tolerance ε	4 mm (laser 1), 6 mm (laser 2)
CAD job file format	DXF
Raspberry Pi–STM32F4 link	UART, 115200 baud, 8N1

Each laser was commanded to move its line by 40, 80, 160, and 320 mm relative to its current position, for example 160 mm to the right, so the recorded error characterizes the motion itself and not the absolute home position. For each laser, displacement, and mode, 15 runs were recorded in three trials of five tests, which gives 240 runs in total. For every run, the camera-measured signed error after the first move, the final signed error, and the number of moves were recorded; in open loop, the final error equals the error after the first move. The measurement noise was estimated separately from 30 images of stationary lines.

III. Registration and Controller Design

The elements of the closed loop in Fig. 3 are described below.

A. Camera-to-Workspace Registration

The workspace is modeled as a plane Π with metric coordinates $\mathbf{x} = [x,\, y]^\top$ defined by the CAD frame of the calibration object. For a pinhole camera, the image coordinates $\mathbf{u} = [u,\, v]^\top$ of a point on Π are related to its workspace coordinates by a planar homography $\mathbf{H} \in \mathbb{R}^{3\times 3}$ [12],

$$\sigma \begin{bmatrix} \mathbf{x} \\ 1 \end{bmatrix} = \mathbf{H} \begin{bmatrix} \mathbf{u} \\ 1 \end{bmatrix}, \tag{1}$$

where σ is a scale factor. The resulting mapping is denoted $\mathbf{x} = h(\mathbf{u}; \mathbf{H})$; lens distortion was not corrected in the prototype.

The calibration object provides N feature points $\mathbf{c}_j$, its corners, whose workspace coordinates are known from its CAD model. Their image locations $\mathbf{u}_j$ are detected and matched to the corresponding CAD features, and $\mathbf{H}$ is estimated by the normalized direct linear transform [12]. For the $N = 4$ corners of the rectangular plate, no three of which are collinear, this estimate is exact; when more features are available, it is

Manuscript 91 submitted to 2026 14th International Conference on Systems and Control (ICSC). Received September 29, 2026.

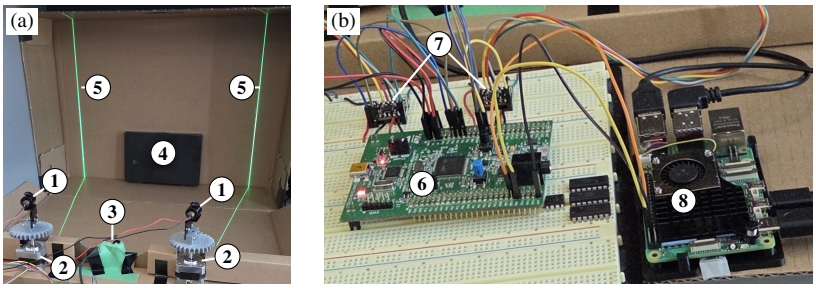


Fig. 2. Experimental prototype. (a) Workspace: (1) green line lasers on (2) NEMA-17 stepper motors, driven directly (the gear-shaped hubs do not mesh with any other gear); (3) camera; (4) calibration plate, 207 × 134 × 6 mm; (5) projected lines on the back panel. (b) Electronics: (6) STM32F4 board, (7) stepper drivers, (8) Raspberry Pi 5.

refined by minimizing the transfer error

$$\hat{\mathbf{H}} = \arg\min_{\mathbf{H}} \sum_{j=1}^{N} \left\| \mathbf{c}_j - h(\mathbf{u}_j; \mathbf{H}) \right\|^2. \tag{2}$$

Because four correspondences determine $\hat{\mathbf{H}}$ exactly, the residual at the corners is zero by construction and does not quantify the registration accuracy elsewhere on Π; this is discussed in Section IV-E.

B. CAD Job Registration and Target Generation

The CAD job file specifies the workpiece contour and a set of task primitives, namely line segments, circles, or points, in the workpiece CAD frame. It is stored in the Drawing Exchange Format (DXF) [15]. This format was chosen because it is openly documented and supported by most CAD packages, represents line, arc, circle, and point entities directly, carries explicit drawing units and named layers that keep the workpiece contour and the different task features apart, and is accepted as input by commercial laser projection systems [16], [17].

The pose of the workpiece on Π is obtained by detecting its contour features $\mathbf{w}_k$ in the image, mapping them to the workspace as $\mathbf{x}_k = h(\mathbf{w}_k; \hat{\mathbf{H}})$, and estimating the planar rigid transformation to the corresponding CAD features $\mathbf{p}_k$ in closed form [18],

$$(\hat{\mathbf{R}}, \hat{\mathbf{t}}) = \arg\min_{\mathbf{R} \in SO(2),\, \mathbf{t} \in \mathbb{R}^2} \sum_{k} \left\| \mathbf{x}_k - (\mathbf{R}\mathbf{p}_k + \mathbf{t}) \right\|^2. \tag{3}$$

Each task primitive is transferred to the workspace by $(\hat{\mathbf{R}}, \hat{\mathbf{t}})$ and into the image by $\hat{\mathbf{H}}^{-1}$, which defines the target used by the controller and a region of interest for detecting the corresponding laser pattern. In the experiments, no separate workpiece was used, so (3) was not required: the target lines were specified in the DXF job file on the wall, on both sides of the calibration object and outside its area (Fig. 1), at the commanded displacements from the start position of each line.

C. Laser Pattern Measurement

The green laser lines are segmented by color within the region of interest of each target, and the detected stripe points are mapped to the workspace by $\hat{\mathbf{H}}$; the dedicated subpixel stripe-center extraction methods of structured-light sensing [19], [20] are not required, since the stripe only has to be located well within the tolerance. Measuring in workspace rather than image coordinates makes the error directly interpretable in millimeters and, for circular patterns, avoids the bias of perspective projection, under which the image of a circle's center does not coincide with the center of the imaged ellipse.

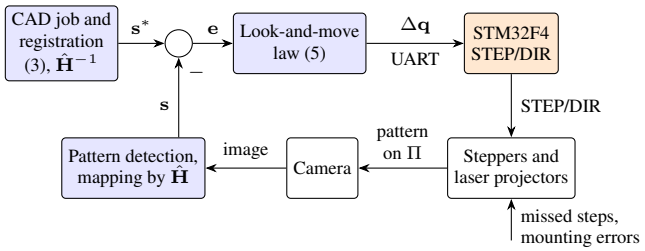


Fig. 3. Closed-loop scheme: blue blocks run on the Raspberry Pi 5, the orange block on the STM32F4; actuator errors act inside the loop.

For a target line $\mathbf{n}_i^{*\top}\mathbf{x} = d_i^*$ with unit normal $\mathbf{n}_i^*$, the positioning error of laser i is the signed perpendicular distance

$$e_i = d_i^* - \mathbf{n}_i^{*\top}\bar{\mathbf{x}}_i, \tag{4}$$

where $\bar{\mathbf{x}}_i$ is the centroid of the detected stripe points. For the vertical target lines of the experiments, $\mathbf{n}_i^*$ is the unit vector of the x axis of Π, the direction of the commanded moves, so a positive error means that the line stopped short of its target and a negative error that it passed it. More generally, the pattern is described by a feature vector $\mathbf{s}_i \in \mathbb{R}^{m_i}$ with target $\mathbf{s}_i^*$ and error $\mathbf{e}_i = \mathbf{s}_i^* - \mathbf{s}_i$, extended by the line angle for a rotational degree of freedom or given by the fitted center for a circle.

D. Closed-Loop Positioning Controller

1) Control Law: Let $\mathbf{q}_i \in \mathbb{Z}^{n_i}$ denote the motor positions of projector i in microsteps. The pattern state depends on $\mathbf{q}_i$ through a map $\mathbf{s}_i = f_i(\mathbf{q}_i)$ determined by the projector optics, mounting, and transmission, whose local sensitivity $\mathbf{J}_i = \partial f_i / \partial \mathbf{q}_i$ is not known accurately. The controller uses a constant estimate $\hat{\mathbf{J}}_i$, i.e., a fixed conversion between millimeters and microsteps that ignores the tangent geometry of a rotary projector. Because images captured during motion are blurred, the controller operates in a look-and-move fashion [6]: the next image is captured only after a settling time T_s following each command, which exceeds the duration of

Manuscript 91 submitted to 2026 14th International Conference on Systems and Control (ICSC). Received September 29, 2026.

the longest move, and the step increment is

$$\Delta\mathbf{q}_i(k) = \mathrm{round}\Big(\mathrm{sat}_{\Delta q_{\max}}\big(\lambda\,\hat{\mathbf{J}}_i^{+}\tilde{\mathbf{e}}_i(k)\big)\Big), \tag{5}$$

where $\tilde{\mathbf{e}}_i(k)$ is the error measured by (4) in image k, $\lambda \in (0,1]$ is the loop gain, $\hat{\mathbf{J}}_i^{+}$ is the pseudo-inverse of $\hat{\mathbf{J}}_i$, $\mathrm{sat}_{\Delta q_{\max}}(\cdot)$ limits each component to $\pm\Delta q_{\max}$, and $\mathrm{round}(\cdot)$ returns the nearest integer number of microsteps. The loop terminates when $\|\tilde{\mathbf{e}}_i(k)\| \le \varepsilon$ in one image. Since each motor affects only its own pattern, the projectors are controlled by independent loops that share each image, so the scheme applies to any number of projectors, provided their regions of interest are disjoint or their patterns are otherwise distinguishable; in the prototype, both lines are green and are assigned to their lasers by disjoint regions of interest (Table I). In the experiments, $\lambda = 1$, and the open-loop reference of Section IV executes only the first move, which is identical to the first closed-loop move.

2) Convergence and Achievable Tolerance: For one feature per laser ($m_i = n_i = 1$) and a locally constant sensitivity J, the index i is omitted. Applying (5) without saturation to the measured error $\tilde{e}(k) = e(k) + \nu(k)$, where $e(k)$ is the true error and $|\nu(k)| \le \bar{\nu}$ the measurement noise, gives

$$e(k+1) = (1-\lambda\rho)\,e(k) - \lambda\rho\,\nu(k) - J\eta(k), \tag{6}$$

where $\rho = J/\hat{J}$ is the sensitivity ratio and $|\eta(k)| \le 1/2$ is the rounding error in microsteps. With $c = |1-\lambda\rho|$, it follows that

$$|e(k+1)| \le c\,|e(k)| + \tfrac{1}{2}|J| + \lambda\rho\,\bar{\nu}. \tag{7}$$

Hence, for $0 < \lambda\rho < 2$ the error contracts until it enters the band $|e| \le b_\infty$, with

$$b_\infty = \frac{|J|/2 + \lambda\rho\,\bar{\nu}}{1-c}. \tag{8}$$

This is the ultimate-boundedness argument of quantized feedback [21], applied to a stepper-actuated projector. Missed steps act as a further bounded disturbance, adding $|J|$ times the largest number of missed microsteps per iteration to the numerator of (8), and saturation only slows the approach to the band.

Two design conclusions follow. First, for any sensitivity error with $0 < \rho < 2/\lambda$, the loop converges to the band $|e| \le b_\infty$: $\hat{J}$ must have the correct sign, but its magnitude may be substantially wrong, for instance because J varies over the range of a rotary projector. Second, termination is guaranteed if $\varepsilon > b_\infty + \bar{\nu}$; for $0 < \lambda\rho \le 1$, (8) gives $b_\infty = |\hat{J}|/(2\lambda) + \bar{\nu}$ and the sufficient condition

$$\varepsilon > \frac{|\hat{J}|}{2\lambda} + 2\bar{\nu}. \tag{9}$$

For $1 < \lambda\rho < 2$, $1 - c = 2 - \lambda\rho$ enlarges b_∞ only slightly while $\lambda\rho$ is close to one. Conversely, the rounding in (5) returns zero steps whenever $\lambda|\tilde{e}| < |\hat{J}|/2$, so for $\varepsilon < |\hat{J}|/(2\lambda)$ the loop stalls regardless of noise. Condition (9) thus bounds the achievable tolerance from below; a small gain, which improves robustness to sensitivity mismatch, raises this bound. Since the bound is dominated by $|\hat{J}|/2$, finer microstepping (up to 256 microsteps per full step on the TMC2209) lowers it proportionally until microstep accuracy under load dominates [5]. On termination, $|e| \le \varepsilon + \bar{\nu}$ with respect to the registered target. For the first move, (6) without noise gives $e(1) = (1-\lambda\rho)\,e(0) - J\eta(0)$: an error proportional to the commanded displacement $e(0)$ plus a rounding offset of at most $|J|/2$ that is fixed for each displacement.

3) Real-Time Step Generation: The STM32F4 receives each command as a pair of signed step increments $(\Delta q_1, \Delta q_2)$ over the UART link and generates the STEP pulses of each axis at up to $f_{\max} \approx 1000$ pulses/s, with the DIR line set before the first pulse. The next image is captured no earlier than $T_s = 1.5$ s after a command, which exceeds the duration of every move in the experiments; the longest, 320 mm, takes about 80 microsteps, i.e., less than 0.1 s at this rate. The duration of one iteration, $T_{\mathrm{it}} \ge T_s + T_{\mathrm{img}}$, where T_{img} covers image acquisition and processing, is therefore dominated by T_s. Steps missed by a motor are not detected by the microcontroller, but they appear as a residual error in the next image and are corrected by the outer loop.

IV. Experimental Results

The system was evaluated with the procedure of Section II-B in terms of convergence, positioning accuracy, and repeatability, and compared with open-loop positioning. For run $r \in \{1,\dots,R\}$, let $e_f^{(r)}$ denote the final error (4) of a laser and $e_1^{(r)}$ its error after the first move, so that $e_f^{(r)} = e_1^{(r)}$ in open loop. The mean absolute error and root mean square error are

$$\mathrm{MAE} = \frac{1}{R}\sum_{r=1}^{R}\big|e_f^{(r)}\big|, \quad \mathrm{RMSE} = \sqrt{\frac{1}{R}\sum_{r=1}^{R}\big(e_f^{(r)}\big)^2}, \tag{10}$$

and are computed analogously for $e_1^{(r)}$. The repeatability σ_{rep} is the standard deviation of the errors over the 15 runs of one laser with the same commanded displacement, pooled as the root mean square over the eight laser–displacement combinations. These errors are evaluated in the camera-registered workspace frame: they quantify how accurately the lasers reach the targets defined through $\hat{\mathbf{H}}$ and include the image measurement noise, but not the registration error, which is estimated in Section IV-D.

A. Closed-Loop Convergence

Fig. 4 shows the absolute error of every closed-loop run before the first move, where it equals the commanded displacement, and after each move. In 100 of the 120 runs, the first move already placed the line within the tolerance. The other 20 runs, ten per laser, needed one correction, with first-move errors of 4.05 to 15.91 mm for laser 1 (all at 320 mm) and 6.56 to 8.43 mm for laser 2 (one trial each at 320 and 40 mm). Each correction reduced the absolute error to at most 1.64 mm, so all runs converged within two moves, 1.17 on average, without stalling or hunting.

These results agree with Section III-D. With 1.8° full steps and four microsteps, one microstep turns a laser by 0.45°

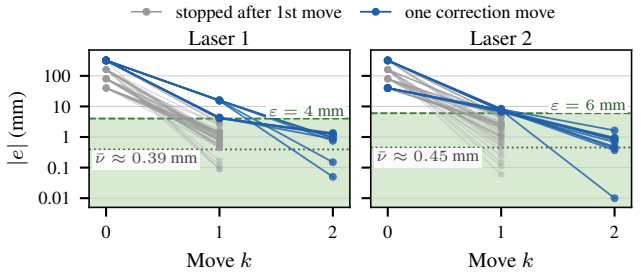


Fig. 4. Measured absolute positioning error of all 120 closed-loop runs versus move number. At $k = 0$, the error equals the commanded displacement. The shaded band marks the tolerance ε, and the dotted line the noise allowance $\bar{\nu} = 3\sigma$.

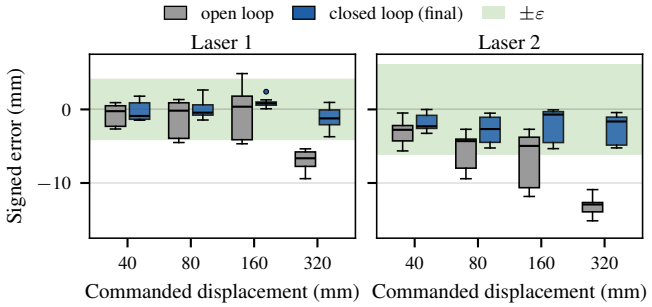


Fig. 5. Measured signed error of the open-loop runs and final signed error of the closed-loop runs for each displacement (15 runs per box). Boxes: median and interquartile range; whiskers: extreme values within 1.5 times that range; dots: values beyond. Shaded: tolerance $\pm\varepsilon$.

and moves its line on the wall, about 500 mm away, by $|J| \approx 3.9$ mm, more at oblique incidence. With $\lambda = 1$, c from the open-loop slopes of Section IV-B (0.022 and 0.034), and the noise of the second measurement, (7) bounds the error after one correction from the largest first-move errors (15.91 and 8.43 mm) by about 3.1 and 3.2 mm, dominated by $|J|/2$; the observed maxima were 1.36 and 1.64 mm. With $\lambda\rho \approx 1.02$–1.03, (8) gives tolerance floors of about 2.9 and 3.0 mm; the tolerances of 4 and 6 mm, chosen from the backlash (lost motion on reversal) of each drive, were 1.4 and 2.0 times higher, and $\varepsilon_1 = 4$ mm is about one microstep on the wall. Image processing times were not recorded separately; with T_s per move, the convergence time was at least 1.75 s on average and 3.0 s at most.

B. Positioning Accuracy and Repeatability

Fig. 5 compares the open-loop errors with the final closed-loop errors for each displacement. In open loop, the error grew with the commanded displacement: the MAE increased from 1.13 mm at 40 mm to 6.88 mm at 320 mm for laser 1, and from 3.15 to 13.19 mm for laser 2. Least-squares fits of the signed open-loop error against the displacement gave slopes of -0.022 and -0.034 ($R^2 = 0.49$ and 0.70), i.e., $\lambda\rho \approx 1.022$ and 1.034 in the registered frame by (6), or 1.035 and 1.047 on the wall (Section IV-D), which the tangent geometry of a rotary projector would produce at incidence angles of about 11° and 12° if $\hat{J}$ is the normal-incidence value. The intercepts, 1.08 and -1.99 mm, may partly stem from rounding $d/\hat{J}$, a fixed offset of up to $|J|/2$ per displacement, but this cannot

TABLE II
MEASURED POSITIONING PERFORMANCE (120 RUNS PER MODE)

Metric	Open loop	Closed loop		Reduction
		1st move	final	vs. open
MAE (mm)	5.10	3.01	1.68	67.1%
RMSE (mm)	6.57	4.53	2.22	66.2%
Max. abs. error (mm)	15.14	15.91	5.33	64.8%
Repeatability σ_{rep} (mm)	2.29	2.75	1.49	34.9%
Moves (mean/max.)	1/1	–	1.17/2	–
RMSE on the wall, est. (mm)	8.63	6.31	3.88	55.0%

explain their variation between trials (per-trial intercepts from -2.7 to 4.1 mm and from -5.3 to -0.1 mm). With feedback, the final MAE no longer depended on the displacement, ranging from 0.85 to 1.43 mm for laser 1 and from 1.84 to 2.77 mm for laser 2. The larger error of laser 2 (RMSE 2.85 mm, versus 1.33 mm for laser 1) follows from its larger tolerance: its first-move errors were all negative, and whenever they fell inside $\varepsilon_2 = 6$ mm the loop stopped without correction, so its final errors retained a mean of -2.26 mm.

The measurement noise had standard deviations of 0.131 and 0.151 mm, giving empirical allowances $\bar{\nu} = 3\sigma \approx 0.39$ and 0.45 mm; they capture jitter but not a possible bias of the stripe detection. The recorded final errors satisfy $|\tilde{e}| \le \varepsilon$ by construction (maxima 3.71 and 5.33 mm), and the true final errors are bounded by $|e| \le \varepsilon + \bar{\nu} \approx 4.39$ and 6.45 mm.

C. Quantitative Performance Comparison

Table II summarizes the three cases; reductions are computed from the displayed values. Since the open-loop move is identical to the first closed-loop move, their RMSE difference (6.57 versus 4.53 mm) reflects variation between trials, and the within-run reductions, 51.0% in RMSE and 66.5% in maximum error, are the cleaner measure of feedback. Relative to the open-loop runs, the registered-frame reductions were 64.8 to 67.1% for the MAE, RMSE, and maximum error and 34.9% for the repeatability.

D. Estimated Accuracy on the Wall

The errors above refer to targets registered on the face of the calibration plate, $\Delta z \approx 6$ mm in front of the wall, with the camera about 500 mm from the wall ($Z_c \approx 494$ mm from the plate face). For a camera facing the wall, a line on the wall then appears on Π with its distance from the foot of the perpendicular from the camera scaled by $Z_c/(Z_c + \Delta z)$, i.e., mislocated by

$$\delta = \frac{d\,\Delta z}{Z_c}, \tag{11}$$

where d is its distance from that foot. Displacements along the wall scale likewise, independent of the start position: a registered displacement d is $(1 + \kappa)d$ on the wall, with $\kappa = \Delta z/Z_c \approx 0.012$, so a run with registered error e and commanded displacement d has the wall-frame error

$$e_{\mathrm{wall}} = (1 + \kappa)\, e - \kappa\, d. \tag{12}$$

The term κd alone is 0.5, 1.0, 1.9, and 3.9 mm at the four displacements, comparable to the registered closed-loop errors

Manuscript 91 submitted to 2026 14th International Conference on Systems and Control (ICSC). Received September 29, 2026.

at 160 mm and larger at 320 mm. Applied to the runs, (12) gives estimated wall-frame RMSEs of 8.63 mm in open loop and 3.88 mm in closed loop (Table II), a reduction of 55.0%. In closed loop, the wall-frame RMSE rises from 2.09 mm at 40 mm to 5.98 mm at 320 mm, where the registration term dominates; the controller drives each line to its registered target as designed. The estimate assumes a plate flush with and parallel to the wall and a camera facing it; registering on the wall, or correcting with the known plate thickness, removes κd.

E. Discussion and Limitations

In the registered frame, the tolerance limited the final error: corrections ended within 1.64 mm, whereas runs that stopped after the first move kept up to 3.71 and 5.33 mm. A tolerance closer to the floor of about 2.9 to 3.0 mm would reduce it at the cost of more moves, and the 1.5 s settling time, which dominated each move, was not optimized.

The following points were not measured and are stated as analytical expectations. First, by Section III-D, the converged error does not depend on projector-model errors or missed steps, so a displaced workpiece or a disturbed laser mount would only require re-registering the workpiece (Section III-B), not updating the projector model; only a camera motion relative to Π, which invalidates $\hat{\mathbf{H}}$, requires recalibration. Second, the registration error at the targets, which lie outside the plate, was not measured beyond the plate offset; it grows with the distance from the corners, over which corner-detection errors are extrapolated, and with radial lens distortion, which was not corrected and is largest near the image edges where the targets lie. Third, detection was evaluated under laboratory lighting on matte cardboard; bright ambient light and reflective metal, typical of welding and cutting and known to disturb stripe detection [20], were not tested. Finally, the benefits for operators (less manual marking; better accuracy, repeatability, and speed) require task-level studies [2], and circles and other patterns remain to be tested.

V. CONCLUSIONS

This paper presented a vision-based closed-loop positioning system in which a DXF job file, registered through a calibration object, defines image-space targets, and a quantized look-and-move law drives stepper-actuated laser projectors until the observed patterns reach them, so that actuator and projector-model errors are corrected by feedback. The analysis showed convergence to a band around the target for $0 < \lambda\rho < 2$ and a lower bound (9) on the achievable tolerance. In 240 runs on a two-laser prototype, every closed-loop run met its tolerance of 4 or 6 mm within two moves, each correction reduced the error to at most 1.64 mm, and feedback reduced the RMSE with respect to the registered targets from 4.53 mm after the first move to 2.22 mm, compared with 6.57 mm in open loop. The 6 mm offset of the calibration plate from the wall adds a registration scale error of about 1.2%, which raises the estimated RMSE on the wall to 3.9 mm in closed loop and 8.6 mm in open loop. Registration on the working surface is therefore a priority for future work, together with finer microstepping, shorter settling times, non-planar and reflective workpieces, other patterns, and operator studies.

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Manuscript 91 submitted to 2026 14th International Conference on Systems and Control (ICSC). Received September 29, 2026.